Ex 1.7:- Diagonal, Triangular and symmetric matrices

Diagonal:- "A diagonal a square matrix where all off the main diagonal are zero." In diagonal 0's can be here.

Upper Triangular:- All elements below the diagonal elements are zeros.

Lower Triangular:- All elements above the " " 0.

- "And if both above and below elements are zero so diagonal matrix."
- A triangular matrix is invertible if and only if its diagonal entries are all non-zero's.

> "For a matrix to be invertible its diagonal entries should not be equal to 0!"

Symmetric:- A square matrix is symmetric if $A = A^t$.

$$\underbrace{(AB)^T = B^TA^T}_{} = BA$$

"Product of a symmetric matrices is symmetric if A and B matrix compute."

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Ex 1.7:- (1-10, 19-28).

Q1) (a). Invertible Upper (c). Diagonal Both (Invertible)

(b). NOT Invertible Lower. (d). NOT Invertible Upper.

Q2) (a). Invertible lower (c). Invertible (Both Diagonal)

(b). NOT Invertible Upper (d). NOT Invertible lower.

Q3).
$$\overset{3\times3}{\begin{bmatrix}3&0&0\\0&-1&0\\0&0&2\end{bmatrix}} \cdot \overset{3\times2}{\begin{bmatrix}2&1\\-4&1\\2&5\end{bmatrix}}$$

Ans:-
$$\begin{bmatrix}6&3\\4&-1\\4&10\end{bmatrix}$$

Q4).
$$\overset{2\times3}{\begin{bmatrix}1&2&-5\\-3&-1&0\end{bmatrix}} \overset{3\times3}{\begin{bmatrix}-4&0&0\\0&3&0\\0&0&2\end{bmatrix}}$$

Ans:-
$$\begin{bmatrix}-4&6&-10\\12&-3&0\end{bmatrix}$$

Q5).
$$\overset{3\times3}{\begin{bmatrix}5&0&0\\0&2&0\\0&0&-3\end{bmatrix}} \overset{3\times5}{\begin{bmatrix}-3&2&0&4&-4\\1&-5&3&0&3\\-6&2&2&2&2\end{bmatrix}}$$

Ans:-
$$\begin{bmatrix}-15&10&0&20&-20\\2&-10&6&0&6\\18&-6&-6&-6&-6\end{bmatrix}$$

Q6).
$$\overset{3\times3}{\begin{bmatrix}2&0&0\\0&-1&0\\0&0&4\end{bmatrix}} \overset{3\times3}{\begin{bmatrix}4&-1&3\\1&2&0\\-5&1&-2\end{bmatrix}} \begin{bmatrix}-3&0&0\\0&5&0\\0&0&2\end{bmatrix}$$

Ans:-
$$\begin{bmatrix}8&-2&6\\-1&-2&0\\20&4&-8\end{bmatrix} \begin{bmatrix}-3&0&0\\0&5&0\\0&0&2\end{bmatrix}$$

$$\begin{bmatrix}-24&-10&12\\3&-10&0\\-18&20&-16\end{bmatrix}$$

Q7). **Ans:** A^2:- $\begin{bmatrix}1&0\\0&4\end{bmatrix}$ A^{-4}:- $\begin{bmatrix}1/1^4&0\\0&1/2^4\end{bmatrix}$

A^{-2}:- $\begin{bmatrix}1&0\\0&1/4\end{bmatrix}$

Q8) **Ans:-** $\begin{bmatrix}-6&0&0\\0&3&0\\0&0&5\end{bmatrix} \xrightarrow{A^2} \begin{bmatrix}36&0&0\\0&9&0\\0&0&25\end{bmatrix}$

A^{-2}:- $\begin{bmatrix}1/36&0&0\\0&1/9&0\\0&0&1/25\end{bmatrix}$ A^{-4}:- $\begin{bmatrix}1/(-6)^4&0&0\\0&1/(3)^4&0\\0&0&1/5^4\end{bmatrix}$

Q9). **Ans:-** $\begin{bmatrix}1/2&0&0\\0&1/3&0\\0&0&1/4\end{bmatrix}$

A^2:- $\begin{bmatrix}1/4&0&0\\0&1/9&0\\0&0&1/16\end{bmatrix}$ A^{-2}:- $\begin{bmatrix}4&0&0\\0&9&0\\0&0&16\end{bmatrix}$

A^{-4}:- $\begin{bmatrix}1/(1/2)^4&0&0\\0&(1/(1/3)^4)&0\\0&0&1/(1/4)^4\end{bmatrix}$

Q10) **Ans:-** $\begin{bmatrix}-2&0&0&0\\0&-4&0&0\\0&0&-3&0\\0&0&0&2\end{bmatrix}$

A^2:- $\begin{bmatrix}4&0&0&0\\0&16&0&0\\0&0&9&0\\0&0&0&4\end{bmatrix}$ A^{-4}:- $\begin{bmatrix}1/(-2)^4&0&0&0\\0&(1/(-4)^4)&0&0\\0&0&(1/(-3)^4)&0\\0&0&0&(1/2^4)\end{bmatrix}$

A^{-2}: $\begin{bmatrix}1/4&0&0&0\\0&1/16&0&0\\0&0&1/9&0\\0&0&0&1/4\end{bmatrix}$

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Q19). NOT Invertible.

Q20). Invertible

Q21). Invertible

Q22). NOT Invertible.

Q23). 3/6 0 0

Ans:- $\begin{bmatrix} -3 & 16 & 63 \\ 0 & 5 & -3 \\ 0 & 0 & -6 \end{bmatrix}$

so → −3, 5, −6.

Q24)

Ans:- $A = \begin{bmatrix} 4 & 0 & 0 \\ -2 & 0 & 0 \\ -3 & 0 & 7 \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 1 & 5 & 0 \\ 3 & 2 & 6 \end{bmatrix}$

$\begin{bmatrix} 24 & 0 & 0 \\ -12 & 0 & 0 \\ 3 & 14 & 42 \end{bmatrix}$

So → 24, 0, 42

Q25). $\begin{bmatrix} 4 & -3 \\ a+5 & -1 \end{bmatrix} = \begin{bmatrix} 4 & a+5 \\ -3 & -1 \end{bmatrix}$

Ans:-

$-3 = a+5$

$a = -8$

Q26). Ans:- $\begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$

=

$\begin{bmatrix} 2 & 3 & 0 \\ a-2b+2c & 5 & -2 \\ 2a+b+c & a+c & 7 \end{bmatrix}$

$a-2b+2c = 3$, $2a+b+c = 0$

$b+c = -2a$

$b = -2a-c$

$a+c = -2$

$a-2(-2a-c)+2c = 3$ | $3-4c = -10-5c$

$a+4a+2c+2c = 3$ | $c = -13$, $b = 9$

$5a+4c = 3 \rightarrow \frac{3-4c}{5} = -2-c$

Q27). Ans:- $\begin{bmatrix} x-1 & x^2 & x^4 \\ 0 & x+2 & x^3 \\ 0 & 0 & x-4 \end{bmatrix}$

$x \neq 1, x \neq -2, x \neq 4$

Q28) Ans:- $x \neq 1/2, x \neq 1/3, x \neq -1/4$

Ex: 1.8 (1-24, 27-41).

Q1).(a) Domain = 2, codomain = 2,

(b). Domain = 3, codomain = 2

(c) Domain = 3, codomain = 3

(d). Domain = 6, codomain = 1

Note:- In $m \times n$ → (n) domain, (m) codomain

Q3). (a). Domain = 2, codomain = 2

(b) Domain = 2, codomain = 3

Q4) (a) Domain = 3, codomain = 3

(b) Domain = 3, codomain = 2

Q5) (a) Domain = 3, codomain = 2

(b) Domain = 2, 〃 = 3

Q7). (a) 〃 = 2, 〃 = 2

(b). 〃 = 3, 〃 = 2

Q9) D = 2, C.D = 3

Q11) (a).

$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 1 \\ 3 & 5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

↓ standard

(b). $\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -8 \\ 0 & -1 & 5 \\ 4 & 7 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

↓ standard.

Q12 Done

Q13).

(a). $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 \\ x_1+3x_2 \\ x_1-x_2 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ -1 \end{bmatrix} : \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 1 & 3 \\ 1 & -1 \end{bmatrix}$ matrix.

(b). $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \\ -7 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 7 & 2 & -1 & 1 \\ 0 & 1 & 2 & 0 \\ -7 & 0 & 0 & 0 \end{bmatrix}$ → Matrix

Q14). (a). $T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ → matrix

(c). $\begin{bmatrix} x_1+2x_2+x_3 \\ x_1+5x_2 \\ x_3 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ matrix.

Q15) Ans:-

$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 3 & 5 & -1 \\ 4 & -1 & 1 \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$

$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -3+10-4 \\ -4-2+4 \\ -3+4-4 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ -3 \end{bmatrix}$

Q16). Done.

Q17) (a). $x = (-1, 4)$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_1+x_2 \\ x_2 \end{bmatrix}$

$= \begin{bmatrix} 5 \\ 4 \end{bmatrix}$

Q19) (a). [2x2] [2x1]

Ans:- $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

(b). 2x3 3x1

Ans:- $\begin{bmatrix} -1 & 2 & 0 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$

$\begin{bmatrix} 3 \\ 13 \end{bmatrix}$

Q21 (a). $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x+y \\ x-y \end{bmatrix}$

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